

To find the number of unordered partitions, we have to count the ----- partitions and then divide it by suitable number to erase the order in partitions.

Answer

☐ unordered

☒ ordered

☐ random

☐ None of these

The greatest common divisor of 5 and 10 is

Answer

☒ 5

☐ 0

☐ 1

☐ None of these

What is the smallest integer N such that $\lceil N/5 \rceil = 7$?

Answer

☒ 35

☐ 31

☐ 30

☐ 0

Compute $\lfloor -1.01 \rfloor$

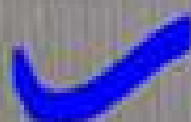
Answer

-2

-1

0

1



The of the experiment is the set of possible outcomes.

Answer

☐ event

☒ sample space

☐ subset

The expectation μ for the following table is

x_i	1	3
$f(x_i)$	0.4	0.1

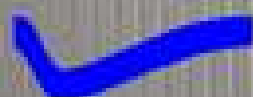
Answer

☐ 0.5

☐ 3.4

☐ 0.3

☒ 0.7



Two distinct edges with the same set of end points are called

Answer

☐ Isolated

☐ Incident

☒ Parallel

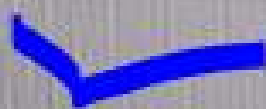
A walk that starts and ends at the same vertex is called

Answer

Simple walk

Circuit

Closed walk



A matrix in which number of rows and columns are equal is called

Answer

☐ Rectangular Matrix

☒ Square Matrix

☐ Null Matrix

☐ Row Matrix

The list of the degrees of the vertices of a graph in non-increasing order is called

Answer

Isomorphic Invariant

Degree Sequence



Order of Graph

Length of Circuit

A vertex of degree 1 in a tree is called a

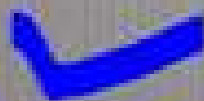
Answer

☐ Internal Vertex

☐ Ancestor

☐ Forest

☒ Terminal vertex



Suppose that a connected planar simple graph has 30 edges. If a plane drawing of this graph has 20 faces, how many vertices does the graph have?

Answer

12



13

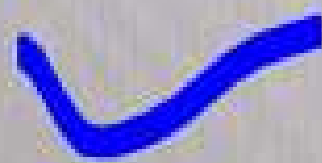
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A graph that consist of a single vertex is called

Answer

Trivial tree



Empty tree

Forest tree

Binary tree

A sub-graph of a graph G that contains every vertex of G is a tree and specifically called

Answer

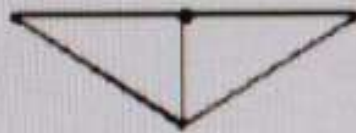
☐ Trivial tree

☐ Empty tree

☒ Spanning tree

☐ Forest

How many non-isomorphic spanning trees does the following simple graph has?



Answer

☒ 2



☐ 4

☐ 6

☐ 8

The graph G is a complete and " n " represents the no. of vertices. Then G is planar if

Answer

☐ $n = 4$

☐ $n < 4$

☐ $n > 4$

☒ $n \leq 4$

If A, B and C are any three events, then $P(A \cup B \cup C)$ is equal to

Answer

☐ $P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C)$

☒ $P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$

☐ $P(A) + P(B) + P(C) + P(A \cap B \cap C)$

☐ $P(A) + P(B) + P(C)$

The probability of getting a 5 when a die is thrown will be

Answer

$$\frac{1}{3}$$

$$\frac{5}{6}$$

$$\frac{1}{6}$$



$$\frac{1}{36}$$

If A and B are independent events then $P(A|B) =$

Answer

$$P(A \cup B)$$

$$P(A \cap B)$$

$$P(A) \quad \checkmark$$

$$P(B)$$

Length of the walk of a graph is

Answer



The number of edges in walk



Total number of edges in a graph



Total number of vertices in a graph



The number of vertices in walk

Suppose that there are eight runners in a race first will get gold medal, the second will get silver and third will get bronze. How many different ways are there to award these medals if all possible outcomes of race can occur and there is no tie.

I

Answer

☐ $P(100,97)$

☐ None of these

☒ $P(8,3)$

☐ $P(97,3)$

The probability of getting 2 heads in two successive tosses of a balanced coin is

Answer

☒ $\frac{1}{4}$

☐ $\frac{1}{3}$

☐ $\frac{1}{2}$

☐ $\frac{2}{3}$

If a die is thrown then the probability that the dots on the top are prime numbers or odd numbers is

Answer

☐ $\frac{1}{3}$

☒ $\frac{2}{3}$

☐ 1

☐ $\frac{1}{2}$

Quotient Remainder Theorem states that for any positive integer d , there exist unique integer q and r such that $n = d \cdot q + r$ and

Answer

None of these

$0 < r < d$

$0 \leq r < d$ ✓

$0 \leq d < r$

If $n(A_i)$ denotes the number of elements in set A_i then

$$n(A_1 \cup A_2 \cup \dots \cup A_m) = n(A_1) + n(A_2) + \dots + n(A_m) \text{ where } \dots$$

Answer

$$A_i \cup A_j = \emptyset, \text{ if } i = j$$

$$A_i \cap A_j = \emptyset, \text{ if } i \neq j$$



$$A_i \cap A_j = \emptyset, \text{ if } i = j$$

$$A_i \cup A_j = \emptyset, \text{ if } i \neq j$$

If one event can occur in n_1 ways and if for each of these n_1 ways, a second event can occur in n_2 ways, then the total number of ways in which both events occur is

Answer

$n_1 + n_2$

$n_1 - n_2$ ✓

$2n_1 n_2$

$n_1 \cdot n_2$

The value of $\frac{(n+1)!}{(n-1)!}$ is

Answer

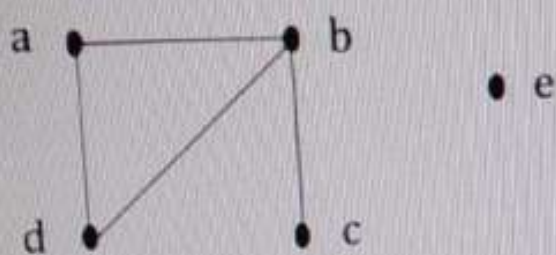
☐ cannot be determined

☒ $n^2 + n$ ✓

☐ 0

☐ $n - 1$

The degrees of the vertices a, b, c, d and e respectively, in the given graph are



Answer

2, 3, 1, 0, 1

2, 3, 1, 2, 0



2, 2, 3, 1, 1

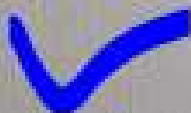
0, 1, 2, 2, 0

An integer n is odd for some integer k iff

Answer

$$n = 2(k + 1)$$

$$n = 2(k - 1)$$

$$n = 2k + 1$$
 

$$n = 2k$$

The product AB of two matrices A and B is defined, if the

Answer

number of columns of $A =$ number of columns of B

number of columns of $B =$ number of rows of A

number of rows of $A =$ number of rows of B

number of columns of $A =$ number of rows of B ✓

A function that is called by itself, either directly or indirectly is called as

Answer

☐ Super function

☒ Recursive function

☐ Main function

☐ All of them

Let n and d be the integers and $d \neq 0$. Then n is divisible by d or d divides n iff

Answer

☐ $n, d = 1$

☐ $n - k = d$

☐ $n, k = 0$

☒ $n = k, d$ for some integer k

In probability distribution random variable f satisfies the conditions

Answer

☐ $f(x_i) < 0$ and $\sum_{i=0}^n f(x_i) = 1$

☐ $f(x_i) \geq 0$ and $\sum_{i=0}^n f(x_i) \neq 1$

☒ $f(x_i) \geq 0$ and $\sum_{i=0}^n f(x_i) = 1$ ✓

☐ $f(x_i) \leq 0$ and $\sum_{i=0}^n f(x_i) \neq 1$

An arrangement of objects with the consideration of order is called _____.

Answer

permutation



combination

selection

sample

If the order matters and repetition is allowed then total number of ways for selecting k sample from n number of elements is _____

Answer

$C(n + k - 1, k)$

$P(n, k)$

$C(n, k)$

n^k



An arrangement of objects without the consideration of order is called _____.

Answer

permutation

combination



sample

probability

If A and B are finite (overlapping) sets, then which of the following must be true?

Answer

☒ $n(A \cup B) = n(A) + n(B) - n(A \cap B)$ ✓

☐ $n(A \cup B) = n(A) - n(B)$

☐ $n(A \cup B) = n(A) + n(B)$

☐ $n(A \cup B) = n(A) + n(B) + n(A \cap B)$

What is the contra positive of the given statement:

If square root of every prime number is irrational then square root of 2 is irrational.

Answer

✓ If square root of 2 is not irrational then square root of every prime number is not irrational.

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If square root of every prime number is not irrational then square root of 2 is not irrational.